



Foreground/Background-Masked Interaction Learning for Spatio-temporal Action Detection

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ABSTRACT

Spatio-temporal Action Detection (SAD) aims to recognize the multi-class actions, and meanwhile locate their spatio-temporal occurrence in untrimmed videos. Besides relying on the inherent inter-actor interactions, most previous SAD approaches model actor interactions between multi-actors and the whole frames or special parts (e.g., objects/hands). However, such approaches are relatively graceless by 1) roughly treating all various actors to equivalently interact with frames/parts or by 2) sumptuously borrowing multiple costly detectors to acquire the special parts. To solve the above dilemma, we propose a novel Foreground/Background-masked Interaction Learning (dubbed as FBI Learning) framework to learn the multi-actor features by attentively interacting with the hands-down foreground and background frames. Specifically, we first design a new Mask-guided Cross Attention (MCA) mechanism that calculates the masked cross-attentions to capture the compact relations between the actors and foreground/background regions. Next, we present a new Actor-guided Feature Aggregation (AFA) scheme that integrates foreground- and background-interacted actor features with the learnable actor-based weights. Finally, we construct a long-term feature bank that associates temporal context information to facilitate action classification. Extensive experiments are conducted on commonly available UCF101-24, MultiSports, and AVA v2.1/v2.2 datasets, which illustrate the competitive performance of FBI Learning against the state-of-the-art methods.

CCS CONCEPTS

• Computing methodologies → Activity recognition and understanding.

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KEYWORDS

video spatio-temporal action detection, 3d convolutional neural network, mask-guided attention, actor-guided feature aggregation

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1 INTRODUCTION

Spatio-temporal Action Detection (SAD) aims to locate the action's temporal and spatial occurrence while recognizing its multiple classes in untrimmed videos [1, 54, 57]. It has broad application prospects, such as autonomous driving, video surveillance, human-computer interaction, and so on [4, 72]. Generally, temporal modeling can capture the temporal information of the inter-frame, which is beneficial for SAD [3, 9, 11, 43]. Since actors' actions always occur in interaction with their surrounding environment, it has recently become a common paradigm to explore spatial interactions based on the completion of temporal modeling to additionally capture the spatial relations of the intra-frame [8, 17, 31, 33, 41, 49, 56, 58, 59, 70]. To explore the spatial relations of actors, a straightforward approach is to model the direct interactions between actors and the whole frame (including objects, persons, environment, etc.), besides the inherent inter-actor interaction [33, 61, 62], as shown in the top left of Figure 1. However, this type of approach cannot accurately model the detailed interaction actions at such a coarse-grained level. Another type of approach models the interactions between actors and the special parts, such as hands or objects, at a fine-grained level [8, 31, 52], as shown in the bottom left of Figure 1. This type of approach has to borrow multiple costly detectors sumptuously, e.g., hand detectors, object detectors, pose estimators, etc; it also needs to further traverse all detected parts, which is extremely tedious. Intrinsically, actors sometimes interact with nearby objects almost in the area where actors are clustered, while sometimes interacting with distant objects in the remaining area.

Formally, we define the area where actors are clustered as the foreground and the remaining area as the background. This means that the foreground contains fine-grained interaction information

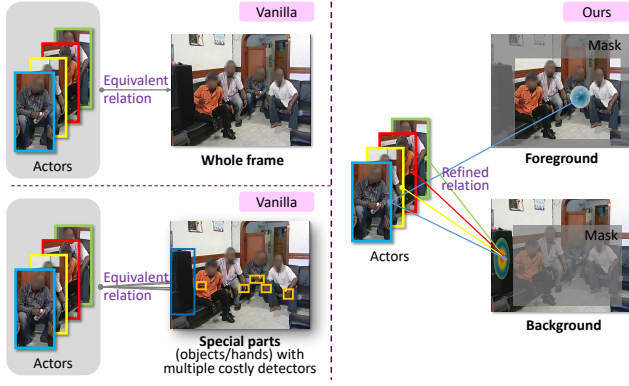


Figure 1: Main insight of our FBI Learning. Besides the inter-actor relation, (i) vanilla approaches treat all actors in an identical manner, and compute their ergodic relation with the whole frame or special parts acquired by multiple costly detectors; and (ii) FBI Learning especially considers the action difference among actors, and captures their compact relations with foreground and background without extra detectors.

between actors and nearby objects, while the background can provide potential interaction information between actors and distant objects from a comprehensive view. We expediently generate the foreground and background via masks based on the location of the actors. As shown in the right of Figure 1, when an actor performs "holding sth.", the foreground information (e.g., hand, and object) becomes relatively critical, while when she/he performs "watching TV", the background information (e.g., television) takes on greater significance. In cases where an actor performs both actions simultaneously, we attempt to balance foreground and background information by weighting them accordingly.

Based on this, we propose a novel Foreground/Background-Masked Interaction Learning (FBI Learning) framework to learn multi-actor features by facilitating the attentive interactions between actors and foreground/background frames without any costly detectors, as shown in Figure 2. Specifically, we design a new Mask-guided Cross Attention (MCA) mechanism to learn the actor features interacted with the foreground/background-masked features by computing foreground- and background-masked cross-attention maps that reflect the actor-background interactions and actor-foreground attentions, respectively. After MCA, we obtain two-stream actor features brought in the actor-background relation and actor-foreground relation, respectively. Subsequently, we present a new Actor-guided Feature Aggregation (AFA) scheme to aggregate the two-stream actor features with the learnable weight factors into the final action features. In addition, to enhance the temporal modeling capability, inspired by [33, 61], we construct a feature bank based on the action features received from AFA, which enriches the temporal contextual connections of actors. Finally, we conduct extensive experiments to validate the performance of the proposed FBI Learning compared with the state-of-the-art methods on the densely annotated datasets (i.e., UCF101-24 [46], Multi-Sports [24]), and the sparsely annotated dataset (i.e., Atomic Visual Actions (AVA) dataset [14]) for the SAD task.

Overall, the main contributions of this work are as follows: (1) We propose a novel Foreground/Background-masked Interaction Learning (FBI Learning) to address the problem of Spatio-temporal Action Detection (SAD) by exploring the relations between actors and foreground/background. (2) To capture the compact relations between actors and foreground/background, we design a new Mask-Guided Cross-Attention (MCA) mechanism to interact with the information between the foreground/background-masked feature and the actor features by leveraging the foreground/background-masked cross-attentions. (3) We present a new Actor-guided Feature Aggregation (AFA) scheme to aggregate two-stream actor features with the learnable weight factors by considering the actor-foreground relation and actor-background relation corresponding to actors.

2 RELATED WORK

Spatio-temporal action detection. In the early stage, one common solution for SAD adopted a two-stream architecture that employs RGB streams for visual representation, optical streams for motion representation, and deep convolutional neural networks for feature extraction [13, 39, 44, 71]. Subsequently, some methods [3, 11, 15, 19, 28] mainly investigated the backbone designing, and utilized various 3D-CNN structures to extract features with temporal continuity. For example, Feichtenhofer *et al.* [11] proposed a two-branch network to capture the spatio-temporal information when extracting action features. Furthermore, some approaches [8, 33, 52, 65–67] focused on uncovering spatial relations in videos, and explicitly modeling the interactions between actors and related objects/environment. Besides the spatial relation, Wu *et al.* [61] proposed a long temporal dependency model to explore the temporal relation in videos. Compared with the above two-stage methods, the one-stage methods [4, 22, 48] usually utilized a one-stage network to facilitate end-to-end training of actor localization and action classification.

Interaction for action detection. Temporal interaction modeling involves entity (e.g., actors, object) interactions [30, 32, 40, 42, 61, 74] at different frames or time intervals. Specifically, some works [30, 74] concentrated on examining temporal relations within video clips to analyze the interactions between adjacent frames. Conversely, the other works [61, 62] aimed to combine information from different clips within the same video and detect action at the broadest possible time scale. Spatial interaction modeling focuses on entity interactions at different spatial locations within the same frame or temporal segment [18, 31, 48, 50, 58, 63, 70]. For example, Sun *et al.* [48] proposed an actor-centric relation network to build an interaction network between the actor and the surrounding background. Furthermore, some works [33, 62] sought to analyze the interactions between actors and the entire scene rather than the surrounding background. To capture the fine-grained interactions, some works [8, 31, 52] incorporated additional special parts (i.e., objects, hands, and pose estimation), and modeled the interactions between actors and the special parts for improving the performance of SAD.

Mask attention. Mask attention is initially introduced in the field of NLP to mask the irrelevant parts of a sentence after padding [55] or to prevent label leakage [21, 51]. With the emergence of Vision Transformer [6], attention masks are gradually becoming widely

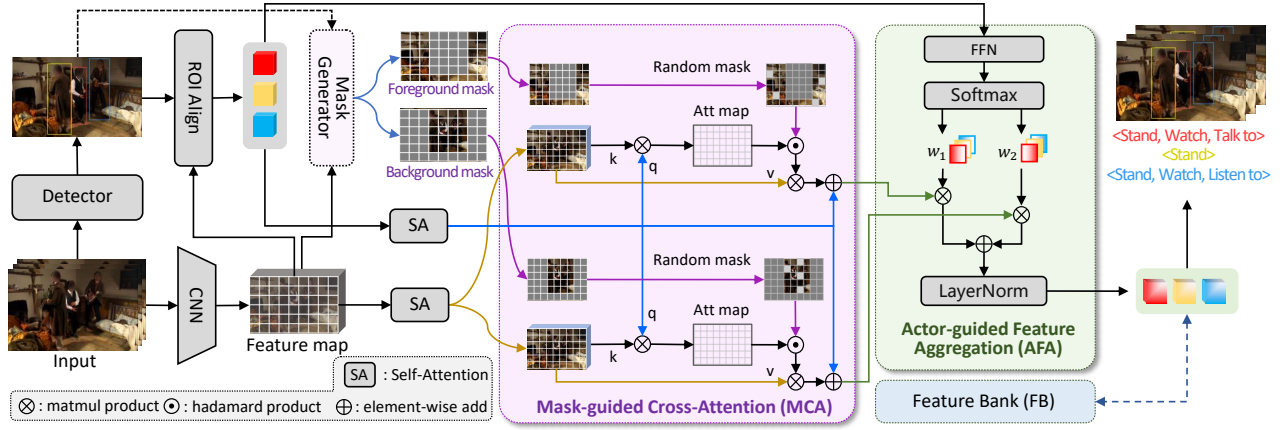


Figure 2: Framework of FBI Learning. Given the input video clip, we adopt an off-the-shelf person detector to generate the actor proposals, employ CNN to produce the video feature, and further, the actor features are obtained via ROI Align. Based on the feature map and actor proposals, the foreground and background masks are generated via Mask Generator, respectively. MCA simultaneously performs the interactions between the actor features and foreground/background-masked video features to obtain two-stream interacted actor features. These two-stream interacted actor features are aggregated with the learnable weight factors (w_1 and w_2) via AFA for predicting the actions.

used in the field of multimedia and computer vision [5, 16, 34, 60]. For example, in the image segmentation task, Cheng *et al.* [5] employed an attention mask in the transformer decoder to restrict attention to the local features centered on the predicted segment. In the human-object interaction detection task, Zhong *et al.* [73] proposed to mask the top scores in cross-attention maps of the decoder layer to construct hard-positive queries for improving the model robustness. Moreover, the integration of mask and attention has gained popularity in the paradigm of self-supervised learning [10, 23, 53]. These techniques aim to extract semantic information from images or videos by utilizing mask reconstruction. In particular, Feichtenhofer *et al.* [10] and Tong *et al.* [53] proposed to mask a significant portion of the video in space and time to capture the robust spatio-temporal characteristics of the footage.

3 FRAMEWORK

3.1 Overview

The framework of FBI Learning is shown in Figure 2, which mainly consists of Mask Generator, MCA and AFA. First, given an input video clip $V_{in} \in \mathbb{R}^{C \times T \times H \times W}$, where C , T , H , and W represent the color channel, number of frames, height, and width, respectively, FBI Learning extracts the corresponding video features using a backbone. Here, we perform adaptive average pooling along the temporal dimension to reduce computational overhead and obtain the feature map $F_v \in \mathbb{R}^{C' \times H' \times W'}$. Simultaneously, we generate actor proposals using an off-the-shelf person detector on the video clip’s keyframe, and then apply the 7×7 RoI Align [16] followed by an adaptive max pooling to obtain N actor features $\{a^i \in \mathbb{R}^{C'}\}_{i=1}^N$, where N indicates the number of actors. Second, the feature map $F_v \in \mathbb{R}^{C' \times H' \times W'}$ and the locations of actor proposals are fed into the Mask Generator (see Section 3.2) to produce the foreground mask and background mask. Third, by taking the actor features $\{a^i\}_{i=1}^N$ and video feature map F_v refined by self-attention as the

query and key/value inputs, the MCA (see Section 3.3) calculates the foreground/background-masked cross-attention maps to generate two-stream actor features, namely the actor-foreground interaction features $\{f_{fore}^i \in \mathbb{R}^{C'}\}_{i=1}^N$, and actor-background interaction features $\{f_{back}^i \in \mathbb{R}^{C'}\}_{i=1}^N$. Fourth, the AFA (see Section 3.4) aggregates such two-stream actor features with learnable weight factors (w_1 and w_2 , based on the actor features $\{a^i\}_{i=1}^N$) into the short-term features $\{f_s^i \in \mathbb{R}^{C'}\}_{i=1}^N$. Meanwhile, we archive all the short-term features to construct Feature Bank $\mathcal{B}$. For the current timestamp t of the video, we retrieve all M archived features from $\mathcal{B}$ in the time window $[t - c, t + c]$, denote as $\{f_1^j\}_{j=1}^M$. Finally, we aggregate $\{f_s^i\}_{i=1}^N$ and $\{f_1^j\}_{j=1}^M$ (see Section 3.5) to detect actions of the N actors at timestamp t , and train a new complete FBI Learning model with the support of the Feature Bank for action detection.

3.2 Mask Generator

The mask generator utilizes the actor proposals to determine the mask area. As discussed in Section 3.1, similar to state-of-the-art methods [11, 33, 52, 61], the actor proposals are obtained using an off-the-shelf person detector (e.g., Faster-RCNN [37], YOLO [36]). Specifically, we select the middle frame of a video clip as the key frame to generate N actor proposals $B_a \in \mathbb{R}^{N \times 4}$, and replicate them across the adjacent frames. We obtain the foreground area R_{fore} by taking the union of multiple actor nominations or expanding a single actor nomination with a certain factor. Correspondingly, we refer to the complement of the foreground on the entire scene as the background area R_{back} . This allows the foreground area to contain information about nearby objects, while the background area can supplement potential information about distant objects. We provide the computational definition of R_{fore} below:

$$R_{fore} = \begin{cases} \text{Union}(B_a) & N > 1, \\ \text{Extend}(B_a, \alpha) & N = 1, \end{cases} \quad (1)$$

where $\text{Union}(\cdot)$ takes the maximum range of the point set, while $\text{Extend}(\cdot, \alpha)$ extends the range of the point set by a factor of α .

To attentively model the interactions between actors and the foreground/background, it's necessary to mask the rest of the scene (background/foreground), which we refer to as the background mask M_{back} and foreground mask M_{fore} . For clarity, we describe the generation process of the foreground mask. First, we use an all-one binary matrix that is consistent with the spatial dimension of the video feature map F_v as the initial mask $M \in \mathbb{R}^{H' \times W'}$. We then perform a mask operation on the spatial positions in the mapped area R_{fore} in the initial mask matrix, setting the values to $-\infty$ with a high ratio β_{fore} , which generates the foreground target mask T_{fore} . Second, we assign a low ratio δ_{fore} to the part outside the mapped area R_{fore} , which results in the foreground random mask D_{fore} . Here, the target mask T_{fore} enables the model to capture the relations between actors and the background, while the random mask D_{fore} further improves the robustness for mining semantic information. Finally, we combine the masks of these two features to obtain the final foreground mask M_{fore} :

$$M_{\text{fore}} \leftarrow T_{\text{fore}} + D_{\text{fore}}. \quad (2)$$

Similarly, the background mask M_{back} is generated using the same process, but with the parameters replaced by R_{back} , β_{back} , and δ_{back} , namely $M_{\text{back}} \leftarrow T_{\text{back}} + D_{\text{back}}$. Following this approach, we can achieve better modeling of the interactions between actors and the foreground/background, allowing for improved accuracy and robustness in our model, which has been validated in Table 4(b).

3.3 Mask-guided Cross-Attention (MCA)

MCA aims to capture the relations between actors and foreground/background, guided by the foreground/background-masked cross-attention maps. MCA differs from the typical transformer-based methods, such as those proposed in [2, 72], in that we avoid initializing the query with random embedding. Previous research has shown that the weights of randomly embedded learnable queries during inference are fixed, which leads to two issues: 1) a randomly embedded learnable query can only roughly attend to a certain region of the image [2, 35]; 2) a randomly embedded learnable query is sensitive to the changing of object position during inference [73]. Those directly result in poor detection performance when facing large-range motions in SAD. Instead, we extract the actor features $\{a^i\}_{i=1}^N$ from the video using ROI Align [16] as the query input Q of MCA, allowing for their updating during both training and inference. Specifically, each layer of the encoder is made up of a Self-Attention (SA) layer and layer normalization. The main processing of the query Q input can be summarised in the following equation:

$$Q = \text{LayerNorm}(\text{SA}(A) + A), \quad (3)$$

$$\text{SA}(A) = \text{softmax}\left(\frac{w_q(A) \times w_k(A)^T}{\sqrt{C'}}\right) \times w_v(A), \quad (4)$$

$$w_*(\cdot) = \text{Linear}(\cdot), \quad (5)$$

where $*$ = {q, k, v}, $A = \{a^i \in \mathbb{R}^{C'}\}_{i=1}^N$ denotes the set of actor features, and $Q = \{q^i \in \mathbb{R}^{C'}\}_{i=1}^N$ denotes the set of query input of MCA. In Section 3.1, we discussed that in MCA, the Key/Value (denoted as K/V) refers to the whole feature map generated by the

encoder. Encoding Key/Value requires preserving spatial information to serve as an interactive global space. To achieve this, each Key/Value encoder layer is enhanced with a Feed-Forward Network (FFN), and position embedding is incorporated into $w(\cdot)$, in addition to Self-Attention (SA) and layer normalization, as follows,

$$K/V = \text{FFN}(\text{LayerNorm}(\text{SA}(F_v) + F_v)), \quad (6)$$

where $F_v \in \mathbb{R}^{C' \times H' \times W'}$ is the whole feature map, $\text{FFN}(\cdot)$ is a stack of two convolutional layers, and $K/V \in \mathbb{R}^{C' \times H' \times W'}$ denotes the input Key/Value for MCA.

As known, modeling fine-grained interactions between actors and the environment without extra detectors is challenging. Attending to all critical areas in video feature maps directly using attention mechanisms is difficult because actors may perform multiple-semantic actions simultaneously, leading to an overly large retrieval scope. Additionally, overfitting problems may arise from such manipulated initialization of actor queries in later stages of modeling. Although dropout [47] is a common solution that discards some features to avoid overfitting to some extent, it may sacrifice the performance [29].

Inspired by [45, 53, 73], we present the random masks to enhance the robustness for mining semantic information by randomly masking some information. Specifically, MCA is motivated by two intuitions with two masks: 1) utilizing the foreground/background target masks $T_{\text{fore}}/T_{\text{back}}$ to model actor-foreground/background interactions reducing the noise of the explored interactions without multiple costly detectors; 2) further utilizing foreground/background random masks $D_{\text{fore}}/D_{\text{back}}$ to enhance query robustness for adapting diverse actions.

Formally, by taking $Q = \{q^i\}_{i=1}^N, K, V, M_{\text{fore}}, M_{\text{back}}$ as the inputs, MCA outputs the actor-foreground interaction features $\{f_{\text{fore}}^i\}_{i=1}^N$ and actor-background interaction features $\{f_{\text{back}}^i\}_{i=1}^N$. The MCA is composed of multiple layers, each of which employs Cross-Attention (CA) with a mask operation. Here, $M_{\text{fore}}/M_{\text{back}}$ is applied to the attention map by the Hadamard product to obtain the masked attention map. Taking the actor-foreground interaction features $F_{\text{fore}} = \{f_{\text{fore}}^i\}_{i=1}^N$ of each actor as an example, the overall process can be summarized as:

$$H_{\text{fore}} = \text{softmax}\left(\frac{w_q(Q) \times w_k(K)^T}{\sqrt{C'}}\right) \odot M_{\text{back}} \times w_v(V), \quad (7)$$

$$F_{\text{fore}} = \text{FFN}(\text{LayerNorm}(H_{\text{fore}} + Q)), \quad (8)$$

where $\odot$ denotes Hadamard product.

3.4 Actor-guided Feature Aggregation (AFA)

After MCA, for one actor, there are two-stream actor features (f_{fore}^i and f_{back}^i) that interact with foreground and background, respectively. Actually, some actors mainly interact with the foreground, while some actors mainly interact with the background. Thus, before aggregating two-stream actor features, we should consider the interaction degree between each actor and the foreground/background.

Motivated by this, AFA is designed to aggregate two-stream actor features by using the learnable weight factors based on the difference of actor features a^i . Specifically, We employ a Feed-Forward Network (FFN) followed by a softmax function to compute two

types of weight factors (denoted as w_1^i and w_2^i) for actor feature a^i . Then, two-stream actor features (f_{fore}^i and f_{back}^i) are aggregated into the interaction feature f_s^i based on these weight factors, as follows,

$$w_1^i, w_2^i = \text{softmax}(\text{FFN}(a^i)), \quad (9)$$

$$f_s^i = \text{LayerNorm}(w_1^i \times f_{\text{fore}}^i + w_2^i \times f_{\text{back}}^i), \quad (10)$$

where $\text{LayerNorm}(\cdot)$ denotes the normalization operation.

3.5 Action Classifier

Simple action classifier. Since the localization of actors is accomplished by an off-the-shelf person detector, our work focuses on the classification task of actions. After obtaining the short-term features $\{f_s^i\}_{i=1}^N$ for all actors, we can achieve the action classification via a simple linear function, i.e.,

$$\hat{Y} = \text{Linear}(\{f_s^i\}_{i=1}^N), \quad (11)$$

where $\hat{Y} \in \mathbb{R}^{N \times L}$ denotes the confidence score of L possible actions of N actors.

Action classifier with feature bank. To explore long-range temporal information for action detection, we also construct a Feature Bank [33, 61, 72] that archives short-term interaction features received from AFA at different timestamps of the video. Specifically, we first train a separate FBI Learning model via Simple Action Classifier described above. The short-term features f_s^i of each actor in all clips of the entire video are archived in the Feature Bank. When retraining a new complete FBI Learning with the support of Feature Bank, for the current video clip of the timestamp t , we retrieve all M archived interaction features $\{f_1^j \in \mathbb{R}^C\}_{j=1}^M$ within the time window $[t - c, t + c]$ from the Feature Bank. We then adopt Cross-Attention (CA) to fuse the short-term features $F_s = \{f_s^i\}_{i=1}^N$ and the long-term ones $F_l = \{f_1^j\}_{j=1}^M$, i.e.,

$$F_{\text{sl}} = \text{softmax}\left(\frac{w_q(F_s) \times w_k(F_l)^T}{\sqrt{C'}}\right) \times w_v(F_l), \quad (12)$$

where $F_{\text{sl}} = \{f_{\text{sl}}^i \in \mathbb{R}^{C'}\}_{i=1}^N$ denotes the final fused features. After that, we apply a simple linear classifier as Equation 11 but replace $\{f_s^i\}_{i=1}^N$ with $\{f_{\text{sl}}^i\}_{i=1}^N$ to implement action classification.

4 EXPERIMENTS

4.1 Dataset

UCF101-24. As a subset of UCF101 [46], it contains 24 sport action classes and provides the bounding box for each frame. Following the evaluation settings of previous works [8, 22, 33], we report the performance on split-1 with the metrics of Frame-mAP@IoU=0.5 and Video-mAP at various IoUs.

MultiSports [24]. It utilizes collective sports as the background of the dataset and gathers 247 high-resolution videos from four types of sports: soccer, basketball, volleyball, and aerobics. Following the benchmark [24], due to the long-tailed distribution in the dataset, we conduct experiments with a set of 24,999 instances covering 66 action classes (18,422 for training, 6,577 for validating), and evaluate 60 action classes with the metrics of Frame-mAP@IoU=0.5 and Video-mAP at various IoUs.

AVA v2.1/v2.2 [14]. It contains 430 videos sampled from YouTube covering 80 action classes. Each video has 15 minutes and is annotated in 1-second intervals. Following the benchmark [14], we conduct experiments on 299 videos (234 for training, 64 for validating) with the annotation version v2.1/v2.2, and evaluate 60 action classes with Frame-mAP@IoU=0.5 as the metric.

4.2 Implementation Details

Person detector. For MultiSports and AVA, we employ person detection boxes following [24, 61], both of which are generated by Faster R-CNN [37] with a ResNeXt-101-FPN [26, 64] backbone. The person detector is pre-trained on ImageNet [38] and COCO keypoints [27] and then fine-tuned on the training sets of the AVA and MultiSports datasets. For UCF101-24, we utilize the person detection boxes from [22], which can be regarded as detected by YOLO v2 [36].

Backbone. We utilize the SlowFast networks [11] as the backbone, and augment the spatial resolution of the Res5 by $2 \times$ following [33]. We use the SlowFast-R101 (R50) pre-trained on the Kinetics-700 (400) dataset with a speed ratio $\alpha = 4$, channel ratio $\beta = \frac{1}{8}$, and a temporal stride $\tau = 8$. Thus, when we take 64 frames as clip input, the Slow pathway samples 8 ($T = 8$) frames at a temporal stride of 8 ($\tau = 8$) while Fast pathway samples 32 frames.

Training and inference. We use SlowFast-R50 as the backbone for the UCF101-24 dataset, as well as use SlowFast-R101 for the AVA and MultiSports datasets. For all datasets, α , β_{fore} , β_{back} , δ_{fore} , δ_{back} , and c are set to 2.0, 0.9, 1.0, 0.2, 0.2 and 30, respectively. The Self-Attention (SA) and MCA contain 3 blocks. We optimize all parameters via Stochastic Gradient Descent (SGD) with Nesterov momentum 0.9 and 1 epoch learning rate warm-up. On UCF101-24/MultiSports/AVA, the batch size, initial learning rate, weight decay, and total epoch are set as 12/6/2, 0.000375/0.001875/0.012, $0.1 \times e^{-5}/0.1 \times e^{-3}/0.1 \times e^{-5}$, and 4/6/8, respectively. The learning rate is reduced by a factor of 10 in epoch 2 and 4 on MultiSports, as well as in epoch 4 and 6 on AVA. During inference, we utilize the detected person bounding boxes from [24], [22] and [61] for MultiSports, UCF101-24 and AVA, respectively. We conduct all experiments using 2 GPUs (RTX-Titan).

4.3 Comparison with State-of-the-arts

Comparison on UCF101-24. Table 1 shows the performance comparisons in terms of Frame-mAP@IoU=0.5, Video-mAP@IoU=0.2, and Video-mAP@IoU=0.5 on UCF101-24. These comparisons across three types of metrics consistently demonstrate that FBI Learning achieves better performances against existing methods [8, 11, 33] under the same backbone (i.e., SlowFast-R50, SF-R50 in short). In particular, FBI Learning achieves 86.0 (Frame-mAP) and 69.1 (Video-mAP@IoU=0.5), even better than 81.3 (Frame-mAP) and 60.2 (Video-mAP@IoU=0.5) of TubeR [72] using two-stream inputs. The largest improvement of FBI Learning on UCF101-24 within a great deal of interactions between actors and environments demonstrates that FBI Learning can effectively model the interactions between actors and the environments.

Comparison on MultiSports. Table 1 summarizes the performance comparisons in terms of Frame-mAP@IoU=0.5, Video-mAP@IoU=0.2, and Video-mAP@IoU=0.5 on MultiSports. We find that

Table 1: Performance comparison on UCF101-24 and MultiSports. R and F indicate the RGB input and optical flow one, respectively. SF denotes the SlowFast network. F@0.5 denotes the Frame-mAP@IoU=0.5, and V@0.2 (0.5) denotes the Video-mAP@IoU=0.2 (0.5).

Method	UCF101-24					MultiSports				
	Backbone	Inputs	F@0.5	V@0.2	V@0.5	Backbone	Inputs	F@0.5	V@0.2	V@0.5
ROAD [44]	VGG16	R+F	-	73.5	46.3	SSD	R	3.9	0.0	0.0
ACT [20]	SSD	R+F	67.1	77.2	51.4	SSD	-	-	-	-
YOWO [22]	ResNext-101	R	80.4	75.8	48.8	ResNext-101	R	9.3	10.9	0.9
MOC (K=7) [25]	DLA34	R+F	78.0	82.8	53.8	DLA34	R	22.5	12.1	0.8
MOC (K=11) [25]	DLA34	-	-	-	-	DLA34	R	25.2	12.9	0.6
SF [11]	SF-R50	R	82.4	-	-	SF-R101	R	27.7	24.2	9.6
ACAR [33]	SF-R50	R	84.3	-	-	-	-	-	-	-
TubeR [72]	CSN-152	R	83.2	82.3	58.4	-	-	-	-	-
TubeR [72]	I3D	R+F	81.3	85.3	60.2	-	-	-	-	-
HIT [8]	SF-R50	R	84.8	-	-	SF-R101	R	33.3	27.8	8.8
Ours	SF-R50	R	86.0	86.9	69.1	SF-R101	R	40.8	30.3	9.9

Table 2: Performance comparison on AVA v2.1. FB indicates Feature Bank. K and IG denote the Kinetics and the IG-65M datasets, respectively. SF denotes the SlowFast network. The letters in the Inputs column indicate the input type: R denotes RGB frame, and F denotes optical flow.

Method	Backbone	Pre-trained	Inputs	FB	mAP
<i>One-stage methods</i>					
I3D [3]	I3D-VGG	K400	R+F	×	15.6
ACRN [48]	S3D-G	K400	R+F	×	17.4
STEP [68]	I3D	K400	R+F	×	18.6
VTR [12]	I3D-VGG	K400	R	×	25.2
WOO [4]	SF-R101	K600	R	×	28.0
YOWO [22]	ResNext-101	K400	R	✓	19.2
TubeR [72]	SF-R101	K700	R	✓	31.6
<i>Two-stage methods</i>					
X3D-XL [9]	X3D-XL	K400	R	×	26.1
SlowFast [11]	SF-R101	K600	R	×	26.3
LFB [61]	I3D-101-NL	K400	R	✓	27.7
ACAR [33]	SF-R101	K700	R	✓	30.0
AIA [52]	SF-R101	K700	R	✓	31.2
Ours	SF-R101	K700	R	×	29.8
Ours	SF-R101	K700	R	✓	31.7

FBI Learning achieves consistent performance improvements over alternatives. Unlike UCF101-24, which relies on characteristic visual backgrounds for coarse-grained action detection, MultiSports has a similar background in the same sport, thus providing non-discriminative background information for fine-grained action detection. Thus, traditional methods, e.g., MOC [25], fail to achieve satisfactory results, which directly model the interactions between actors and the entire frame. Due to the consideration of actor-foreground interactions, FBI Learning is capable to capture more reliable relations that are crucial for action detection.

Comparison on AVA v2.1. Table 2 shows the performance comparisons in terms of Frame-mAP@IoU=0.5 on AVA v2.1. In general, the performance of using Feature Bank or not, with the same backbone (i.e., SlowFast-R101, SF-R101 in short), consistently demonstrates that the proposed FBI Learning outperforms existing methods, including both one-stage methods (e.g., WOO [4], TubeR [72])

Table 3: Performance comparison on AVA v2.2. The letters in parentheses following the method name indicate the extra detector utilized (marked in blue): O, H, and P denotes the object detector, hand detector, and pose estimator, respectively.

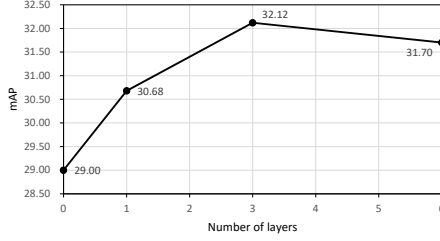
Method	Backbone	Pre-trained	FB	mAP
X3D-XL [9]	X3D-XL	K600	×	27.4
WOO [4]	SF-R101	K600	×	28.3
MVIT-B-24 [7]	MVIT-B-24	K600	×	28.7
SF-101,NL [11]	SF-R101	K600	×	29.0
MAR (O) [69]	SF-R101	K600	×	30.2
TubeR [72]	CSN-152	IG+K400	×	31.1
ACAR [33]	SF-R101	K700	×	31.7
Ours	SF-R101	K700	×	32.1
YOWO [22]	ResNext-101	K400	✓	20.2
AIA (O) [52]	SF-R101	K700	✓	32.2
HIT (O, H, P) [8]	SF-R101	K700	✓	32.6
ACAR [33]	SF-R101	K700	✓	33.3
TubeR [72]	CSN-152	IG+K400	✓	33.6
Ours	SF-R101	K700	✓	33.9

and two-stage methods (e.g., SlowFast [11], ACAR [33], AIA [52]). Specifically, without Feature Bank, FBI Learning outperforms the base backbone SlowFast [11] by 3.5% and the end-to-end method (i.e., WOO [4]) by 1.8%. Alternatively, compared with previous methods with Feature Bank, FBI Learning is also more competitive. Among two-stage methods, FBI Learning performs better than ACAR [33] which also focuses on interaction modeling, demonstrating the superiority of FBI Learning in terms of actor-foreground/background interaction modeling.

Comparison on AVA v2.2. Table 3 shows the performance comparisons in terms of Frame-mAP@IoU=0.5 on AVA v2.2. It's impressive to find that FBI Learning outperforms all comparative methods either with or without Feature Bank, and even significantly surpasses HIT [8] using multiple costly detectors. In particular, without Feature Bank, FBI Learning achieves the performance gains, e.g., 1.0% of gain for TubeR [72] (one-stage method with strong backbones) and 1.9% of gain for MAR [69] (two-stage method with additional detectors). When using the same backbone (i.e., SlowFast-R101),

Table 4: Ablations studies for FBI Learning on AVA v2.2. mAP denotes Frame-mAP@IoU=0.5.

(a) With different components.		(b) With different masks.			(c) With different aggregation strategies.	
Method	mAP	Target Masks (T_{fore}/T_{back})	Random Masks (D_{fore}/D_{back})	mAP	# of Aggregation strategy	mAP
B0 (Baseline)	29.00	×	×	30.75	Sum	31.10
B1 (+CA)	30.75	✓	×	31.88	Concat	31.25
B2 (+MCA)	31.25	✓	✓	32.12	AFM	31.65
B3 (+MCA+AFA)	32.12				AFA	32.12
B4 (+MCA+AFA+FB)	33.93					

**Figure 3: Performance achieved by FBI Learning with the different number of attention layers.**

FBI Learning gains 3.1% and 0.4% compared with SlowFast [11] and ACAR [33], respectively. This proves that the superior performance of FBI Learning owes to its design rather than the backbone.

4.4 Ablation Studies

We conduct all ablation studies on AVA v2.2, and evaluate 60 action classes with Frame-mAP@IoU=0.5 as the metric.

Effect of main components. Table 4(a) shows the performance contribution of each component in FBI Learning. Here, CA indicates MCA without masks. We use a single-classification-head SlowFast-R101 as the Baseline (B0). B0 directly trains the action classifier by using actor features obtained via ROI Align [16] on feature maps with actor proposals, without additional interaction modeling processes. Next, we extend B0 into B1 (+CA) for modeling interactions between actors and the environment via CA, which yields performance improvement. Compared with vanilla Cross-Attention (CA), which roughly integrates all environmental information, MCA captures specific relations between actors and foreground/background. Thus, B2 (+MCA) adopts MCA instead of CA to model actor-foreground/background interactions, and outperforms B1 with traditional Cross-Attention (CA). Furthermore, after adding AFA that effectively aggregates the two-stream actor features generated, B3 (+MCA+AFA) performs better than B2. Ultimately, like most previous works [8, 33, 61, 72] with the support of Feature Bank, B4 (+MCA+AFA+FB, i.e., FBI Learning) continuously gains performance improvement.

Effect of foreground and background masks. To demonstrate the effect of the target masks T_{fore}/T_{back} , as well as the random masks D_{fore}/D_{back} , we conduct ablation studies to evaluate their effectiveness. Table 4(b) shows the performance of FBI Learning with different masks. We observe that using the target masks during training and inference achieves a performance improvement of 1.13% over that without using any mask. It is proved that interactions manipulated by the target masks can capture the relations between actors and foreground/background attentively. Furthermore,

when adding both the target and random masks, the performance of FBI Learning is further improved. It is demonstrated that the random masks can enhance the robustness of mining the semantic information.

Effect of different aggregation strategies. As a new feature aggregation strategy, AFA is deeply investigated compared with Sum aggregation (Sum), Concatenation aggregation (Concat), and Attention Feature Module (AFM) [8]. Here, AFM concatenates the two-stream features and then refines features by the non-local mechanism, which is suitable for merging equal-weighted features. Table 4(c) shows the performance of FBI Learning with different aggregation strategies. We can see that FBI Learning with the Sum aggregation and Concat aggregation are comparable, both of which perform worse than AFM. This is because that AFM considers the differences between foreground and background features. Over AFM, AFA improves the performance since the latter further considers the bias between actor-foreground interaction and actor-background interaction for one specific actor, as well as the relation among the actor features. Specifically, AFA learns two weight factors based on the relation among actor features to aggregate two-stream actor features, which quantificationally considers some actors interacted with the foreground while others may with the background.

Effect of the number of attention layers. MCA is the main component in FBI Learning, thus the number of its attention layers is important. Following [6], we also set the number $a = \{0, 1, 3, 6\}$ in MCA. Figure 3 summarizes the effect of the number a of attention layers in MCA. We can see that FBI Learning achieves the best performance when the $a = 3$, namely 3-layer attention layers in MCA. Notably, when the number of layers is set to 0, the model no longer utilizes cross-attention. In the conventional Transformers, the attention generally involves 6 layers. We speculate that MCA no longer requires such a deep structural layer due to the mirror structure used in MCA.

4.5 Visualization

Visualization of SAD obtained by FBI Learning. Figure 4 shows some results of SAD obtained by FBI Learning. Here, GT refers to ground-truth class labels for actions, Att represents the heat map incorporating the foreground cross-attention map and background cross-attention map, and DET represents the detection results. We can find that the higher-light areas in Att are more attentive, which is consistent with the detection results in DET. In Scene 1 and 2, FBI Learning accurately identifies the actions of both target and non-target actors provided by the offline actor detector, demonstrating its ability to avoid the nonexistent actor-actor relation in



Figure 4: Visualization of SAD obtained by FBI Learning. GT refers to ground-truth class labels for actions, Att represents the heat map incorporating the foreground and background cross-attention map, and DET represents the detection results. In Att, the higher-light areas are more attentive, which is consistent with the detection results in DET.

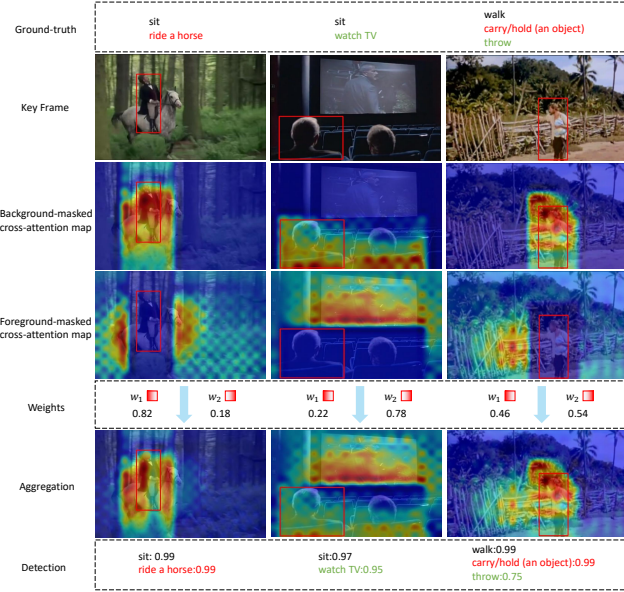


Figure 5: Visualization of masked cross-attention maps. The actions without any interaction are marked in black, the actions of actor-foreground interaction are marked in red, and the actions of actor-background interaction are in green. The corresponding weights by AFA are also given.

some specific cases. In Scene 3 and 4, there are various actor-actor (e.g., "watch (a person)") and actor-object relations (e.g., "torch (an object)"), which are jointly captured by FBI Learning.

Visualization of masked cross-attention maps. Figure 5 visualizes the foreground-masked cross-attention maps and background-masked cross-attention maps in MCA, respectively. It is illustrated

that both foreground- and background-masked cross-attention maps focus on objects strongly related to the action. And then, AFA can effectively combine actor-foreground interactions and actor-background ones by assigning different weights. In some cases, even when one actor interacts with both foreground and background, FBI Learning can still accurately detect the corresponding actions.

5 CONCLUSION

We propose a novel Foreground/Background-masked Interaction Learning (FBI Learning) framework to address the problem of Spatio-temporal Action Detection (SAD). FBI Learning creatively models the attentive interactions between multi-actors and the hands-down foreground/background frames, and thus being superior to vanilla approaches which either treat all actors equivalently interacting with frames/parts or use multiple costly detectors to acquire the special parts. Specifically, FBI Learning provides one solution paradigm that incorporates: 1) Mask Generator aims to produce foreground and background masks; 2) Mask-guided Cross Attention (MCA) aims to generate two-stream actor features by learning actor features interacted with foreground/background-masked features; 3) Actor-guided Feature Aggregation (AFA) aims to aggregate the two-stream actor features with the learnable weight factors into the final action features. Extensive experiments on four datasets demonstrate that the proposed FBI Learning outperforms the state-of-the-art methods.

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REFERENCES

- [1] Liangliang Cao, Zicheng Liu, and Thomas S Huang. 2010. Cross-dataset action detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1998–2005.
- [2] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. 2020. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*. 213–229.
- [3] Joao Carreira and Andrew Zisserman. 2017. Quo vadis, action recognition? a new model and the kinetics dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 6299–6308.
- [4] Shoufa Chen, Peize Sun, Enze Xie, Chongjian Ge, Jiannan Wu, Lan Ma, Jiajun Shen, and Ping Luo. 2021. Watch only once: An end-to-end video action detection framework. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 8178–8187.
- [5] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. 2022. Masked-attention mask transformer for universal image segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1290–1299.
- [6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*. 1–21.
- [7] Haoqi Fan, Bo Xiong, Kartikeya Mangalam, Yanghao Li, Zhicheng Yan, Jitendra Malik, and Christoph Feichtenhofer. 2021. Multiscale vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 6824–6835.
- [8] Gueter Josmy Faure, Min-Hung Chen, and Shang-Hong Lai. 2023. Holistic Interaction Transformer Network for Action Detection. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 3340–3350.
- [9] Christoph Feichtenhofer. 2020. X3d: Expanding architectures for efficient video recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 203–213.
- [10] Christoph Feichtenhofer, Haoqi Fan, Yanghao Li, and Kaiming He. 2022. Masked Autoencoders As Spatiotemporal Learners. *arXiv preprint arXiv:2205.09113* (2022).
- [11] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He. 2019. Slow-fast networks for video recognition. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 6202–6211.
- [12] Rohit Girdhar, Joao Carreira, Carl Doersch, and Andrew Zisserman. 2019. Video action transformer network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 244–253.
- [13] Georgia Gkioxari and Jitendra Malik. 2015. Finding action tubes. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 759–768.
- [14] Chunhui Gu, Chen Sun, David A Ross, Carl Vondrick, Caroline Pantofaru, Yeqing Li, Sudheendra Vijayanarasimhan, George Toderici, Susanna Ricco, Rahul Sukthankar, et al. 2018. Ava: A video dataset of spatio-temporally localized atomic visual actions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 6047–6056.
- [15] Kensho Hara, Hirokatsu Kataoka, and Yutaka Satoh. 2017. Learning spatio-temporal features with 3d residual networks for action recognition. In *IEEE/CVF International Conference on Computer Vision Workshops (ICCV Workshops)*. 3154–3160.
- [16] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. 2017. Mask r-cnn. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2961–2969.
- [17] Jingwei Ji, Ranjay Krishna, Li Fei-Fei, and Juan Carlos Nieves. 2020. Action genome: Actions as compositions of spatio-temporal scene graphs. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10236–10247.
- [18] Jingwei Ji, Ranjay Krishna, Li Fei-Fei, and Juan Carlos Nieves. 2020. Action genome: Actions as compositions of spatio-temporal scene graphs. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10236–10247.
- [19] Shuiwang Ji, Wei Xu, Ming Yang, and Kai Yu. 2012. 3D convolutional neural networks for human action recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35, 1 (2012), 221–231.
- [20] Vicky Kalogeiton, Philippe Weinzaepfel, Vittorio Ferrari, and Cordelia Schmid. 2017. Action tubelet detector for spatio-temporal action localization. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 4405–4413.
- [21] Jacob Devlin Ming-Wei Chang Kenton and Lee Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Annual Conference of the North American Chapter of the Association for Computational Linguistics-Human Language Technologies (NAACL-HLT)*. 4171–4186.
- [22] Okan Köpüklü, Xiangyu Wei, and Gerhard Rigoll. 2019. You only watch once: A unified cnn architecture for real-time spatiotemporal action localization. *arXiv preprint arXiv:1911.06644* (2019).
- [23] Rui Li, Yiting Wang, Wei-Long Zheng, and Bao-Liang Lu. 2022. A Multi-view Spectral-Spatial-Temporal Masked Autoencoder for Decoding Emotions with Self-supervised Learning. In *the 30th ACM International Conference on Multimedia (ACM MM)*. 6–14.
- [24] Yixuan Li, Lei Chen, Runyu He, Zhenzhi Wang, Gangshan Wu, and Limin Wang. 2021. Multisports: A multi-person video dataset of spatio-temporally localized sports actions. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 13536–13545.
- [25] Yixuan Li, Zixu Wang, Limin Wang, and Gangshan Wu. 2020. Actions as moving points. In *European Conference on Computer Vision (ECCV)*. 68–84.
- [26] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. 2017. Feature pyramid networks for object detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2117–2125.
- [27] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *European Conference on Computer Vision (ECCV)*. 740–755.
- [28] Kun Liu, Wu Liu, Chuang Gan, Minghui Tan, and Huadong Ma. 2018. T-C3D: Temporal convolutional 3D network for real-time action recognition. In *AAAI Conference on Artificial Intelligence (AAAI)*. 7138–7145.
- [29] Yiping Lu, Zhuohan Li, Di He, Zhiqing Sun, Bin Dong, Tao Qin, Liwei Wang, and Tie-Yan Liu. 2019. Understanding and improving transformer from a multi-particle dynamic system point of view. *arXiv preprint arXiv:1906.02762* (2019).
- [30] Chih-Yao Ma, Asim Kadav, Iain Melvin, Zsolt Kira, Ghassan AlRegib, and Hans Peter Graf. 2018. Attend and interact: Higher-order object interactions for video understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 6790–6800.
- [31] Joanna Materzynska, Tete Xiao, Roei Herzig, Huijuan Xu, Xiaolong Wang, and Trevor Darrell. 2020. Something-else: Compositional action recognition with spatial-temporal interaction networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1049–1059.
- [32] Jingcheng Ni, Jie Qin, and Di Huang. 2021. Identity-aware Graph Memory Network for Action Detection. In *the 29th ACM International Conference on Multimedia (ACM MM)*. 3437–3445.
- [33] Junting Pan, Siyu Chen, Mike Zheng Shou, Yu Liu, Jing Shao, and Hongsheng Li. 2021. Actor-context-actor relation network for spatio-temporal action localization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 464–474.
- [34] Liang Qiao, Ying Chen, Zhanzhan Cheng, Yunlu Xu, Yi Niu, Shiliang Pu, and Fei Wu. 2021. Mango: A mask attention guided one-stage scene text spotter. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2467–2476.
- [35] Xian Qu, Changxing Ding, Xingao Li, Xubin Zhong, and Dacheng Tao. 2022. Distillation using oracle queries for human-based human-object interaction detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 19558–19567.
- [36] Joseph Redmon and Ali Farhadi. 2017. YOLO9000: better, faster, stronger. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 7263–7271.
- [37] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. 2015. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*. 91–99.
- [38] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. 2015. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision* 115 (2015), 211–252.
- [39] Suman Saha, Gurkirt Singh, Michael Sapienza, Philip HS Torr, and Fabio Cuzzolin. 2016. Deep learning for detecting multiple space-time action tubes in videos. *arXiv preprint arXiv:1608.01529* (2016).
- [40] Xiangbo Shu, Jinhui Tang, Guo-Jun Qi, Wei Liu, and Jian Yang. 2019. Hierarchical long short-term concurrent memory for human interaction recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, 3 (2019), 1110–1118.
- [41] Xiangbo Shu, Jinhui Tang, Guo-Jun Qi, Yan Song, Zechao Li, and Liyan Zhang. 2017. Concurrence-aware long short-term sub-memories for person-person action recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops)*. 1–8.
- [42] Xiangbo Shu, Liyan Zhang, Guo-Jun Qi, Wei Liu, and Jinhui Tang. 2021. Spatiotemporal co-attention recurrent neural networks for human-skeleton motion prediction. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 6 (2021), 3300–3315.
- [43] Xiangbo Shu, Liyan Zhang, Yunlian Sun, and Jinhui Tang. 2020. Host–parasite: Graph LSTM-in-LSTM for group activity recognition. *IEEE Transactions on Neural Networks and Learning Systems* 32, 2 (2020), 663–674.
- [44] Gurkirt Singh, Suman Saha, Michael Sapienza, Philip HS Torr, and Fabio Cuzzolin. 2017. Online real-time multiple spatiotemporal action localisation and prediction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 3637–3646.
- [45] Kaitao Song, Xiu-Shen Wei, Xiangbo Shu, Ren-Jie Song, and Jianfeng Lu. 2020. Bi-modal progressive mask attention for fine-grained recognition. *IEEE Transactions on Image Processing* 29 (2020), 7006–7018.
- [46] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. 2012. UCF101: A dataset of 101 human actions classes from videos in the wild. *arXiv preprint arXiv:1212.0402* (2012).
- [47] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: a simple way to prevent neural networks from overfitting. *The Journal of Machine Learning Research* 15, 1 (2014), 1929–1958.

- [48] Chen Sun, Abhinav Shrivastava, Carl Vondrick, Kevin Murphy, Rahul Sukthankar, and Cordelia Schmid. 2018. Actor-centric relation network. In *European Conference on Computer Vision (ECCV)*. 318–334.
- [49] Chen Sun, Abhinav Shrivastava, Carl Vondrick, Rahul Sukthankar, Kevin Murphy, and Cordelia Schmid. 2019. Relational action forecasting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 273–283.
- [50] Chen Sun, Abhinav Shrivastava, Carl Vondrick, Rahul Sukthankar, Kevin Murphy, and Cordelia Schmid. 2019. Relational action forecasting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 273–283.
- [51] Yu Sun, Shuohuan Wang, Yukun Li, Shikun Feng, Xuyi Chen, Han Zhang, Xin Tian, Danxiang Zhu, Hao Tian, and Hua Wu. 2019. Ernie: Enhanced representation through knowledge integration. *arXiv preprint arXiv:1904.09223* (2019).
- [52] Jiajun Tang, Jin Xia, Xinzhi Mu, Bo Pang, and Cewu Lu. 2020. Asynchronous interaction aggregation for action detection. In *European Conference on Computer Vision (ECCV)*. 71–87.
- [53] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. 2022. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. In *Advances in Neural Information Processing Systems (NeurIPS)*. 1–16.
- [54] Du Tran and Junsong Yuan. 2012. Max-margin structured output regression for spatio-temporal action localization. In *Advances in Neural Information Processing Systems (NeurIPS)*. 1–9.
- [55] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*. 5998–6008.
- [56] Guangzhi Wang, Yangyang Guo, Yongkang Wong, and Mohan Kankanhalli. 2022. Distance Matters in Human-Object Interaction Detection. In *the 29th ACM International Conference on Multimedia (ACM MM)*. 4546–4554.
- [57] Limin Wang, Yu Qiao, and Xiaoou Tang. 2014. Video action detection with relational dynamic-poselets. In *European Conference on Computer Vision (ECCV)*. 565–580.
- [58] Xiaolong Wang, Ross Girshick, Abhinav Gupta, and Kaiming He. 2018. Non-local neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 7794–7803.
- [59] Xiaolong Wang and Abhinav Gupta. 2018. Videos as space-time region graphs. In *European Conference on Computer Vision (ECCV)*. 399–417.
- [60] Yiran Wang, Zhiyu Pan, Xingyi Li, Zhiguo Cao, Ke Xian, and Jianming Zhang. 2022. Less is More: Consistent Video Depth Estimation with Masked Frames Modeling. In *the 30th ACM International Conference on Multimedia (ACM MM)*. 6347–6358.
- [61] Chao-Yuan Wu, Christoph Feichtenhofer, Haoqi Fan, Kaiming He, Philipp Krahenbuhl, and Ross Girshick. 2019. Long-term feature banks for detailed video understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 284–293.
- [62] Jianchao Wu, Zhanghui Kuang, Limin Wang, Wayne Zhang, and Gangshan Wu. 2020. Context-aware rcnn: A baseline for action detection in videos. In *European Conference on Computer Vision (ECCV)*. 440–456.
- [63] Junbin Xiao, Xindi Shang, Xun Yang, Sheng Tang, and Tat-Seng Chua. 2020. Visual relation grounding in videos. In *European Conference on Computer Vision (ECCV)*. 447–464.
- [64] Saining Xie, Ross Girshick, Piotr Dollár, Zhuowen Tu, and Kaiming He. 2017. Aggregated residual transformations for deep neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1492–1500.
- [65] Wentao Xie, Guanghui Ren, and Si Liu. 2020. Video relation detection with trajectory-aware multi-modal features. In *the 28th ACM International Conference on Multimedia (ACM MM)*. 4590–4594.
- [66] Rui Yan, Lingxi Xie, Xiangbo Shu, Liyan Zhang, and Jinhui Tang. 2023. Progressive Instance-Aware Feature Learning for Compositional Action Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (2023), 10317–10330.
- [67] Rui Yan, Lingxi Xie, Jinhui Tang, Xiangbo Shu, and Qi Tian. 2020. HiGCIN: Hierarchical graph-based cross inference network for group activity recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (2020), 6955–6968.
- [68] Xitong Yang, Xiaodong Yang, Ming-Yu Liu, Fanyi Xiao, Larry S Davis, and Jan Kautz. 2019. Step: Spatio-temporal progressive learning for video action detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 264–272.
- [69] Yanyi Zhang, Xinyu Li, and Ivan Marsic. 2021. Multi-label activity recognition using activity-specific features and activity correlations. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14625–14635.
- [70] Yubo Zhang, Pavel Tokmakov, Martial Hebert, and Cordelia Schmid. 2019. A structured model for action detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9975–9984.
- [71] Jiaojiao Zhao and Cees GM Snoek. 2019. Dance with flow: Two-in-one stream action detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9935–9944.
- [72] Jiaojiao Zhao, Yanyi Zhang, Xinyu Li, Hao Chen, Bing Shuai, Mingze Xu, Chunhui Liu, Kaustav Kundu, Yuanjun Xiong, Davide Modolo, et al. 2022. Tuber: Tubelet transformer for video action detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 13598–13607.
- [73] Xubin Zhong, Changxing Ding, Zijian Li, and Shaoli Huang. 2022. Towards Hard-Positive Query Mining for DETR-Based Human-Object Interaction Detection. In *European Conference on Computer Vision (ECCV)*. 444–460.
- [74] Bolei Zhou, Alex Andonian, Aude Oliva, and Antonio Torralba. 2018. Temporal relational reasoning in videos. In *European Conference on Computer Vision (ECCV)*. 803–818.